Dear Editor,

We are pleased to resubmit our revised manuscript entitled *“A 45-Year Global Analysis of the Spatial Human-Forest Nexus”* for your consideration in *Communications Earth & Environment*. We are grateful to the reviewers for their insightful and constructive comments, which have helped us improve the clarity, robustness, and relevance of the manuscript.

This study investigates the evolving spatial relationship between human populations and forest ecosystems over the past 45 years, a critical topic in the context of ongoing deforestation, biodiversity loss, and climate change. To capture these dynamics, we introduce the Forest-Human Nexus (FHN) index—a novel metric integrating forest area per capita, human-forest proximity, and population density—to provide a spatially explicit and temporally consistent measure of human-forest interdependence.

In this revised version, we have made several substantial improvements in response to reviewer feedback:

- Enhanced the analysis of global interactions between forests and human systems, with clearer explanations of the FHN and related indices, and deeper contextualization in the literature.

- Revised the title to more explicitly reflect the spatial dimension of the analysis.

- Strengthened the methodological robustness of the Forest Proximate People indicator, including sensitivity analyses for varying distance thresholds.

- Expanded discussion of regional patterns and their implications, supported by case studies and relevant policy contexts.

We believe that these revisions significantly improve the manuscript and increase its value for readers interested in sustainability, land-use change, and human-environment interactions. We hope that our analysis and the introduction of the FHN index will offer a valuable tool for future research and policy-making aimed at balancing human development with forest conservation.

We appreciate your consideration of our revised manuscript and look forward to your feedback.

Sincerely,

E. Massaro, P. Newton, J. C. Ciscar, G. Dubois, A. Fanelli, D. Ibarreta, N. Riccetti, W. Szewczyk and A. Cescatti